

Ejercicio 5. Obtener un filtro con frecuencia de corte en 1MHz que presente en 1,2MHz una atenuación de 20dB. Calcular la cantidad de celdas necesarias para cumplir con la especificación suponiendo que la celdas son filtros de K constante. Simular el diseño planteado y verificar su funcionamiento.

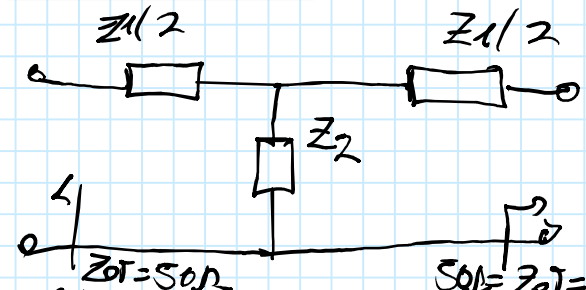
Datos:

$$f_c = 1 \text{ MHz}, \quad \alpha = 20 \text{ dB en } f = 1,2 \text{ MHz} \quad 1 \text{ celda "Klee"}$$

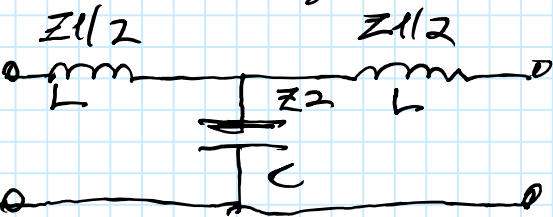
Queremos que atenué en $f > f_c$, unos 20dB que es lo mínimo. Por simplicidad de celdas que tenemos en filtros los pas.

1º) Establezca criterios de diseño $\rightarrow$ celda T. Comenzamos con 1 celda pero vamos viendo cuántas atenua dicho celda y así vamos aumentando celdas del mismo tipo hoy que celdas en cascada.

Celda T en Z



Celda T Paralela en Z



$$Z_{0T} = R \cdot \sqrt{1 + \frac{Z_1^2}{4Z_2}}, \quad R = \sqrt{Z_1 \cdot Z_2}$$

$$\text{We sabemos que } Z_{0T} = 0, \quad Z_1 = j\omega L, \quad Z_2 = \frac{1}{j\omega C}$$

$$Z_{0T} = \sqrt{\frac{1}{L}} \cdot \sqrt{1 + \frac{L^2 C}{4}} = \sqrt{\frac{1}{L}} \cdot \sqrt{1 - \frac{\omega^2 L^2 C}{4}}$$

$$Z_{0T} = \sqrt{\frac{1}{L}} \text{ si } \omega = 0 \text{ con } Z_{0T} = 50 \Omega = 0.6 \text{ dB}$$

• Ahora RL y RC y no son 50 Ohms

$$Z_{0T} = 0 \Rightarrow R \sqrt{1 + \frac{Z_1^2}{4Z_2}} \Rightarrow \left(\frac{0}{R}\right)^2 = \left(1 + \frac{Z_1^2}{4Z_2}\right)^2$$

$$Z_{0T} = 0 \Rightarrow 1 + \frac{Z_1^2}{4Z_2} = 0 \Rightarrow 1 + \frac{j^2 \omega^2 L^2}{4 \cdot \frac{1}{j\omega C}} = 0 \Rightarrow 1 - \frac{\omega^2 L^2 C}{4} = 0$$

$$1 - \frac{\omega^2 L^2 C}{4} = 0 \Rightarrow \frac{\omega^2 L^2 C}{4} = 1 \Rightarrow \omega = \frac{2}{\sqrt{LC}} \quad (1)$$

$$\omega_{0T} = \frac{2}{\sqrt{LC}} = \frac{2}{\sqrt{LC}} = \omega_c$$

$$\text{Por otro lado, } R = \sqrt{L \cdot \frac{1}{C}} = \sqrt{\frac{L}{C}} = R$$

$$Z_{0T} = \sqrt{\frac{1}{L}} \text{ si } \omega = 0 \quad (2)$$

$$\left\{ \begin{array}{l} \omega_c = \frac{2}{\sqrt{LC}} \quad (1) \Rightarrow \sqrt{LC} = \frac{2}{\omega_c} \Rightarrow L = \left(\frac{2}{\omega_c}\right)^2 \cdot \frac{1}{C} \\ Z_{0T} = \sqrt{\frac{1}{L}} \Big|_{\omega=0} = 50 \Omega \quad (2) \Rightarrow L = 50^2 \cdot C \end{array} \right.$$

$$\Rightarrow 50^2 \cdot C = \left(\frac{2}{\omega_c}\right)^2 \cdot \frac{1}{C}$$

$$\Rightarrow C = 6,366 \text{ nF}$$

$$L = 159 \mu\text{H}$$

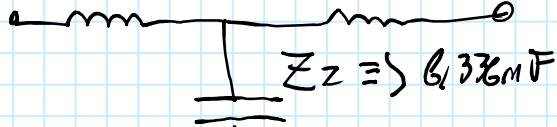
$$I \quad v \sim \quad |w=0$$

$$L = 151 \mu H$$

$$\Rightarrow C = 6,366 mF$$

• 666 T nos quedará como:

$$Z_{12} \Rightarrow 7.9575 \mu H \quad Z_{12} \Rightarrow 7.9575 \mu H$$



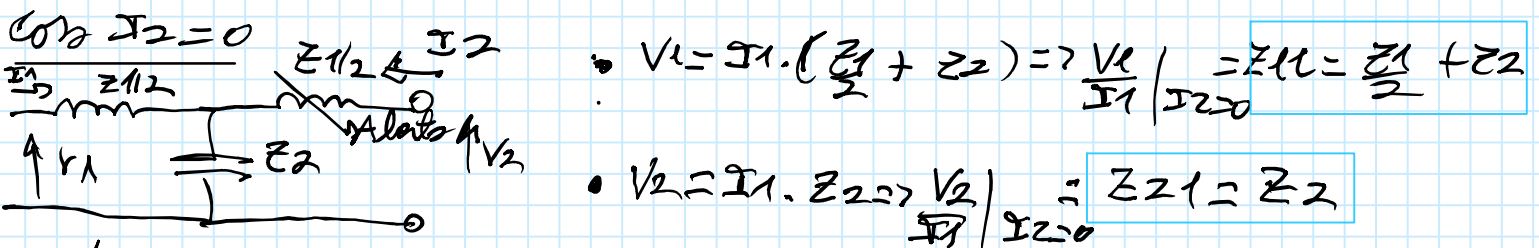
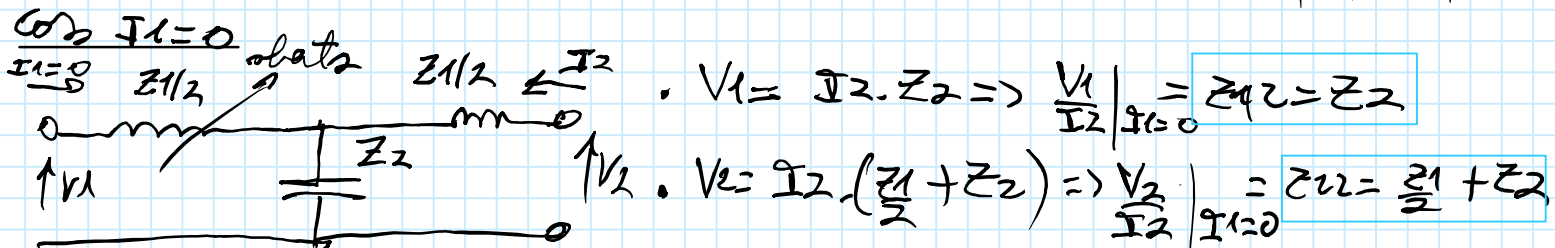
20) Plantea $e^{\alpha + j\omega t} = e^{\alpha} e^{j\omega t}$

• Por otro lado $\Rightarrow e^{\alpha} = e^{\sqrt{A \cdot D} + \sqrt{B \cdot C}}$ y por separación $\Rightarrow A = D$

$$e^{\alpha} = e^{\alpha} e^{j\omega t} = A + \sqrt{B \cdot C} \left\{ \begin{array}{l} |e^{\alpha}| = e^{\alpha} \\ A = \frac{Z_{21}}{Z_{11}}, B = \frac{\Delta Z}{Z_{21}}, C = \frac{1}{Z_{21}} \end{array} \right.$$

• De las ecuaciones paramétricas Z :

$$\begin{aligned} V_1 &= Z_{11} I_1 + Z_{12} I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \Rightarrow \left\{ \begin{array}{l} Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}, Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0} \\ Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}, Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} \end{array} \right.$$



Matriz Z

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_1}{2} + Z_2 & Z_2 \\ Z_2 & \frac{Z_1}{2} + Z_2 \end{bmatrix} = \begin{bmatrix} \frac{\$L}{2} + \frac{1}{\$C} & \frac{1}{\$C} \\ \frac{1}{\$C} & \frac{\$L}{2} + \frac{1}{\$C} \end{bmatrix}$$

$$[Z] = \begin{bmatrix} \frac{\$L + 2}{2\$C} & \frac{1}{\$C} \\ \frac{1}{\$C} & \frac{\$L + 2}{2\$C} \end{bmatrix}$$

$$\begin{array}{r} \frac{1}{2} \\ \frac{1}{2} \\ \hline 1 \end{array} \quad \frac{3x+2}{2x}$$

Ejercicio5_Hoja_3

domingo, 5 de octubre de 2025

18:51

3º) Obtengo A, B, C para plantear e^0

$$A = \frac{Z_{11}}{Z_{21}}, B = \frac{\Delta Z}{Z_{21}}, C = \frac{1}{Z_{21}}$$

$$A = \frac{\$^2 L C + 2}{2 \$ C} = \frac{\$^2 L C + 2}{2} = A$$

$$\Delta Z = Z_{11} \cdot Z_{22} - Z_{12} \cdot Z_{21} = \left(\frac{\$^2 L C + 2}{2 \$ C} \right)^2 - \left(\frac{1}{\$ C} \right)^2 = \frac{\$^4 L^2 C^2 + 4 \$^2 L C + 4}{4 \$^2 C^2} - \frac{1}{\$^2 C^2}$$

$$\Delta Z = (1/4) \frac{\$^4 L^2 C^2 + 4 \$^2 L C + 4}{\$^2 C^2} - \frac{1}{\$^2 C^2} = \frac{\$^4 L^2 C^2 (1/4) + \$^2 L C + 1 - 1}{\$^2 C^2}$$

$$\Delta Z = \cancel{\$^2 C^2} \left[\cancel{\$^2 L^2} (1/4) + \frac{1}{\$ C} \right] = \frac{\$^2 L^2}{4} + \frac{1}{\$ C} = \Delta Z$$

$$B = \frac{\Delta Z}{Z_{21}} = \frac{\frac{\$^2 L^2}{4} + \frac{1}{\$ C}}{\frac{1}{\$ C}} = \frac{\$^3 L^2 C}{4} + \$ L = B$$

$$C = \frac{1}{\$ C} = \$ C = C$$

$$e^0 = \frac{\$^2 L C + 2}{2} + \sqrt{\left(\frac{\$^3 L^2 C}{4} + \$ L \right) \cdot \$ C} = \frac{\$^2 L C + 2}{2} + \sqrt{\frac{\$^4 L^2 C^2}{4} + \$^2 L C}$$

$$\$ = \omega L \Rightarrow e^0 = \frac{-\omega^2 L C + 2}{2} + \sqrt{\frac{\omega^4 L^2 C^2}{4} - \omega^2 L C}$$

$$\text{Para } \omega = 2\pi \cdot 1.2 \mu\text{Hz} = 2.4 \cdot 10^6 \pi \text{ rad/s}$$

$$\frac{-\omega^2 L C + 2}{2} = -1.865 \quad \text{con } L = 15.91 \mu\text{H},$$

$$\Rightarrow e^0 = -1.865 + 1.57 = -0.295$$

$$\sqrt{\frac{\omega^4 L^2 C^2}{4} - \omega^2 L C} = 1.57$$

$$\text{con } L = 15.91 \mu\text{H}$$

$$|e^0| = 0.295 \Rightarrow \ln(0.295) = -1.22 = \alpha \text{ (neper)}$$

$$\alpha \text{ en dB} = -1.22 \cdot 8.686 = -10.6 \text{ dB}$$

1 4

$$\text{con } L = 15,91 \mu\text{H}$$

$$-1,22 \text{ dB (máx)} = \alpha$$

conclusion: Como se puede ver luego 20 dB en $1,2 \text{ MHz} \Rightarrow$ se necesitan
2 elos de $-10,54 \text{ dB}$

Ejercicio5_Hoja_4

martes, 11 de noviembre de 2025

19:10

4to Hemisecación

con $m=0,6$ y $L=15,91 \mu H$, $C=61,366 nF$

